

- 3.7 sin 35° = 2.1 N
horizontally towards the left
- 5 (a) (i) obeys the law $pV/T = \text{constant}$ or any two named gas laws
at all values of p , V and T
or two correct assumptions of kinetic theory of ideal gas
third correct assumption
- (ii) ($pV = nRT$ gives) $(1.00 \times 10^5) (750 \times 10^{-6}) = n (8.31) (300)$
 $n = 0.030$

(b)

	work done on gas / J	heat supplied o gas / J	increase in internal energy of gas / J
A to B	+360	0	+360 &
B to C	0 \$	+670	+670 \$
C to D	-810 &	0	-810
D to A	0 @	-220 @	-220 #

- &: first and third line correct
- \$: second line correct
- #: -220 correct in right hand column
- @: other two figures correct in last line
- (c) the gas molecules bounce off the receding piston at lower speeds
there is a decrease in kinetic energy of the molecules

$$2 \pm \sqrt{2^2(0.002) + 1^2(0.002)}$$